Technical Writing and Presenting

1. Recap and/or Reboot
   - industrial mathematics and computation
   - working in pairs
2. Topics for the first Project
   - proposals of topics
   - deadlines
3. Technical Writing
   - the formal technical report
   - project presentations

MCS 472 Lecture 4
Industrial Math & Computation
Jan Verschelde, 21 January 2026

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industrial mathematics and computation

The three most important aspects of this course:

1. Solving problems in industry, science, and government.
2. Emphasis on projects, technical writing and presenting.
3. Collaborative team work is strongly encouraged.

Homework is meant to explore potential topics for projects and to get familiar with the mathematical and computational methods.

We will have three weeks of project presentations.

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working in pairs

To write programs in pairs, two methodologies can be applied:

1. pair programming

   Two developers share the same computer.
   - The driver codes.
   - The navigator reads and checks the code.

   The roles switch from time to time.

2. code review

   Two developers share their code.
   - The reviewer suggests changes to the code.
   - The coder adjusts the code based on the review.

   The roles switch from time to time.

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1. What is the best CDF for the age of death?

Locate a public database for mortality,
e.g.: the IL department of public health, or census data.

1. Represent the data first as a histogram format, such as the number of deaths in every age group.
2. Construct a polynomial or spline approximation for the CDF.

2. Does the Poisson distribution describe arrivals well?

Gather data about the arrivals (or checkouts) of a store, or alternatively locate a public database for the traffic of a particular web site.

1. Model the probability density function and the cumulative probability distribution.
2. Assuming the distribution is Poisson, what is the value for the lambda parameter?

3. Design an overbooking strategy in an airline.

Suppose each passenger has the same probability p of showing up for the N seats on a flight.

Each empty seat represents a loss of L dollars.

Let m be the number of extra tickets sold.

1. What is the maximum reparation R should be offered to compensate passengers who cannot be given a flight?
2. Use the binomial distribution in your numerical study.

4. What is the expected return of the Illinois lottery?

The prizes and odds of winning in the IL lottery are:

- \$2 million (1 in 15,890,700),
- \$2,000 (1 in 60,192),
- \$50 (1 in 1,120),
- \$5 (1 in 60),
- \$2 (1 in 7.8).

Consider the following questions:

1. What is the justification for these numbers?
2. Assuming one million tickets are sold,
   how large should the prize be before \$1 ticket
   has an expected return of more than \$1.

5. Quality loss of servicing requests.

Long wait times in a store may lead to a loss of customers.

Short checkout times may cause a loss caused by idle cashiers.

1. Run computational experiments to determine the mean and standard deviations
   1. of the wait time of the customers, and
   2. the idle time of the cashiers.
2. Design a quality loss function for this problem.

6. Explore the central limit theorem.

Consider the random variable

$$X = \frac{X_1 + X_2 + \cdots + X_N}{\sqrt{N}}$$

where X_k is the outcome of the k-th flip of a fair coin.

1. Show that $E[X] = 0$ and $E[X^2] = 1$.
2. Show experimentally that for large N, X is normally distributed.

7. Make a Poisson random number generator.

Numbers following the Poisson distribution with a given average number of events can be generated following an algorithm listed in volume 2 of the Art of Computer Programming by Donald E. Knuth.

1. Write an implementation for this algorithm.
2. For sufficiently large experiments, compute the mean and the standard deviation of the generated sequences of numbers.
3. Compare with the Poisson distribution we used in Lecture 3.

8. The Substitute Teacher

Suppose a school has 100 teachers, who teach 100 days
and take one sick day. If all sick days are distinct,
then one substitute teacher suffices to cover all teaching.

1. Formulate this question using the binomial distribution.
   The assumption that all sick days are distinct is unrealistic,
   what is the probability that one substitute will not suffice?
2. Run Monte Carlo simulations to solve this problem.
   Interpret the results of the runs and compare with
   the analysis that applied the binomial distribution.
3. Setup a general model for any number N of teachers,
   any number D of teaching days, with the probability p
   that one teacher reports sick.
   Test your model on the specific $N = 100 = D$, $p = 1/100$.
   Extrapolate for different probabilities p.

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deadlines

1. Commit to a topic by Friday 24 January, 5pm (soft deadline), and at the latest by Monday 27 January, before 5pm.

   Email me before the deadline with your topic and partner.

   If two students choose the same topic, then a collaboration is strongly recommended.

2. Work for two weeks. Start early and work often!

3. Present in the third week, starting Monday 10 February.

4. The technical report is due by 5pm on Friday 14 February.

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the formal technical report

1. The Elements of a Technical Report

- Title and Abstract
- Table of Contents and Nomenclature
- Introduction and Analysis
- Experimental Equipment and Procedure
- Results and Discussion
- Conclusions, References, and Appendices

2.1 Title of the Technical Report

- Consists of seven to eight well-chosen words.
- Goal is to give clear understanding of the content.

2.2 Abstract (Summary)

- Should not be written until all other parts completed.
- Only one or two paragraphs (250-300 words).
- Stated in simple declarative sentences.
- It *must* stand alone from the report.

2.3 Table of Contents

- Lists each section heading along with page numbers.
- Does not include Table of Contents and Abstract.
- Followed by a list of illustrations and a list of tables.

2.4 Nomenclature

- Alphabetic list of all symbols used (first Latin, then Greek).
- Separate list for extensively used acronyms and abbreviations.
- Acronyms and abbreviations must be spelled out at their first appearance, e.g., fast Fourier transform (FFT).

2.5 Introduction encourages to read on

- Describes motivation: why is your work important?
- Provides background: what is the context of your work?
- Ends by describing what follows in the sections.

2.6 Analysis = underlying mathematical basis

- Derivation from well-known basic relationships to the specific formulas used to handle and interpret the data.
- Quote readily available results from the literature.
- Be self-contained, reader should not consult other sources.

2.7 Experimental Equipment and Procedure

- Schematic overview of the experimental equipment, description of algorithms and software used.
- Open to peer review: experiments can be reproduced by someone familiar with the general area.
- Poor quality of tables and pictures may destroy report!
- Every table and picture must have a clear caption, i.e.: legend for column headers and coordinate axes.
- Never include tables or pictures not mentioned in the text.

2.8 Results

- Present answers obtained from experiments, referring to tables and pictures, inform about the analysis of the data.
- Assume some readers will jump immediately to results section.

2.9 Discussion

- Summarizes results, separating the new from the "as expected."
- Compares with similar investigations, pointing out strong points and the limitations of the work.
- Provides interpretation of the results, progressing logically, starting with what is known first, and then what is new second.

2.10 Conclusions

- Concise statements of your results and discussion.
- Clean synthesis without further explanation.

2.11 References

[1] W. Strunk and E.B. White. *The Elements of Style*. Fourth Edition. Longman Publishers, 2000.

2.12 Appendices

- Side issues or (output of) lengthy calculation or codes.

3. Approaching a Writing Task, Style Issues

- Who will read what you are writing?
- Multiple authors: pass text around and discuss your changes to arrive at a neutral, uniform narrative flow.
- Technical writing is direct, concise, and clear.
- Read, revise, read, revise, read, revise...

4. Technical Aspects of Writing

- The typesetting environment:
  - Microsoft products are germane in business and industry.
  - LaTeX is standard for mathematical typesetting.
  - Know how to include graphical results in your documents.
- Writing by hand versus typing on computer:
  - By hand: focus on ideas and content.
  - On computer: fast itemizing, attention for layout.
  - In any case, use of spelling checker is mandatory!

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Project Presentations

1. Do's and Don'ts of general slide presentations:
   1. Why using slides?
   2. What are good slides?
   3. How to present slides well?

2. Essentials of a good project presentation:
   1. accessible to your peers
   2. reflects content of your technical report
   3. some practical details

1.1 Why using slides?

- Secure structure: good slides guide the presenter and prevent omission of important aspects.
- While not so good for mathematical derivations, it is suitable for listing topics and presentation of facts.
- Precise presentation of data, like pictures and tables.

1.2 What are good slides?

- Do not include entire paragraphs, except for quotes.
- Do not overload slides: at most 10 lines per slide.
- Your audience has a limited attention span,
  so make the slides self-contained.
  Do not assume your audience will remember details
  on previous slides.

1.3 How to present slides well?

- Do not read the slides, but tell a story.
  Long lists of items on slides tends to be boring.
- Keep eye contact with your audience.
  Try to tune the pace of your talk to your audience.
  Make your point, slides can *glide* away unnoticed.
- Be mobile and flexible.
  Provide transition sentences between slides.

2.1 What is a good project presentation?

- Know your audience.
  Your talk must be accessible to your peers
- The presentation is independent from the technical report, because your peers have not read the report.
- It is similar to the "executive summary."
  How would you tell the boss about your work in 10 minutes?

2.2 The content of the presentation

- Problem statement and motivation. Be precise.
- What did you do? Why did you do it this way?
- Summary of the data: tables and pictures.
- Results and conclusions.

2.3 Some practical details

- LaTeX or PowerPoint provide uniformity.
- Handwritten slides are flexible.
- Duration: 10 to 15 minutes.
- No more than 8 slides.
- Rehearse with your team members.

bibliography

- The proposals for the topics are adapted from the exercises in the first two chapters of the book of Charles R. MacCluer: *Industrial Mathematics. Modeling in Industry, Science, and Government*. Prentice Hall, 2000.
- The slides on the formal technical report are adapted from Chapter 15 of the above mentioned book,
a chapter written by Craig Gunn and Charles R. MacCluer.